



Single-cell analysis identifies depletion of Antigen-Presenting Cell (APC) and IFN-stimulated CD4 populations associated with high-risk in NDMM patients.



Junia Vieira, MD, PhD¹, Alison S Park, BSc², David Melnekoff, PhD¹, Tarek H. Mouhieddine, MD³, Alessia D'Anna, MSc², Adolfo Aleman, Jr.⁴, Meghana Ram, BSc⁵, Katerina Kappes, BSc⁴, Mark Hamilton, MD⁶, George Mulligan, PhD⁶, Arun P Wiita, MD, PhD⁷, Ajai Chari, MD⁸, Thomas Martin, MD⁹, Michael Slade, MD, MS¹⁰, Li Ding, PhD¹¹, Ravi Vij, MD, MBA¹², Brian Brown, PhD⁵, Benjamin Tycko, MD, PhD¹³, Rena Feinman, PhD¹⁴, David S. Siegel, MD, PhD¹⁵, Sundar Jagannath, MD¹, Samir Parekh, MD¹ and Alessandro Laganà, PhD¹⁶

¹ Tisch Cancer Institute, Icahn School of Medicine at Mount Sinai, New York, NY; ² Icahn School of Medicine at Mount Sinai, New York; ³ Department of Medicine, Hematology and Medical Oncology, Tisch Cancer Institute, Icahn School of Medicine at Mount Sinai, New York, NY; ⁴ Department of Medicine, Hematology and Medical Oncology, Icahn School of Medicine at Mount Sinai, New York, NY; ⁵ Icahn School of Medicine at Mount Sinai, New York, NY; ⁶ Multiple Myeloma Research Foundation, Norwalk, CT; ⁷ Department of Laboratory Medicine, University of California, San Francisco, San Francisco, CA; ⁸ University of California, San Francisco, San Francisco, CA; ⁹ Department of Hematology, University of California at San Francisco, San Francisco, CA; ¹⁰ Division of Oncology, Washington University School of Medicine, Saint Louis, MO; ¹¹ Washington University, St. Louis, MO; ¹² Division of Oncology, Washington University School of Medicine, St. Louis, MO; ¹³ Center for Discovery and Innovation, Hackensack Meridian Health, Nutley, NJ; ¹⁴ Center for Discovery & Innovation, Hackensack Meridian Health, Nutley, NJ; ¹⁵ Hackensack University Medical Center, Hackensack, NJ; ¹⁶ Department of Medicine, Hematology and Medical Oncology, Tisch Cancer Institute, Icahn School of Medicine at Mount Sinai, New York.

BACKGROUND

- Over the past decades, survival in multiple myeloma (MM) has drastically improved due to advances in treatment regimens, including the introduction of immunomodulatory drugs and T cell-mediated immunotherapies.
- The myeloma microenvironment (MM-TME) may have a critical role in disease emergence and treatment outcomes. Therefore, understanding the contributions of the MM-TME in high-risk scenarios is essential to optimize and refine patient risk stratification and management.
- In this study, we integrate single-cell RNA seq (scRNA-seq) with tumor whole exome sequencing (WES) to analyze the MM-TME and tumor compartments in a cohort of newly diagnosed (NDMM) high risk (HR) and standard risk (SR) MM patients.

METHODS

Patient selection and study design

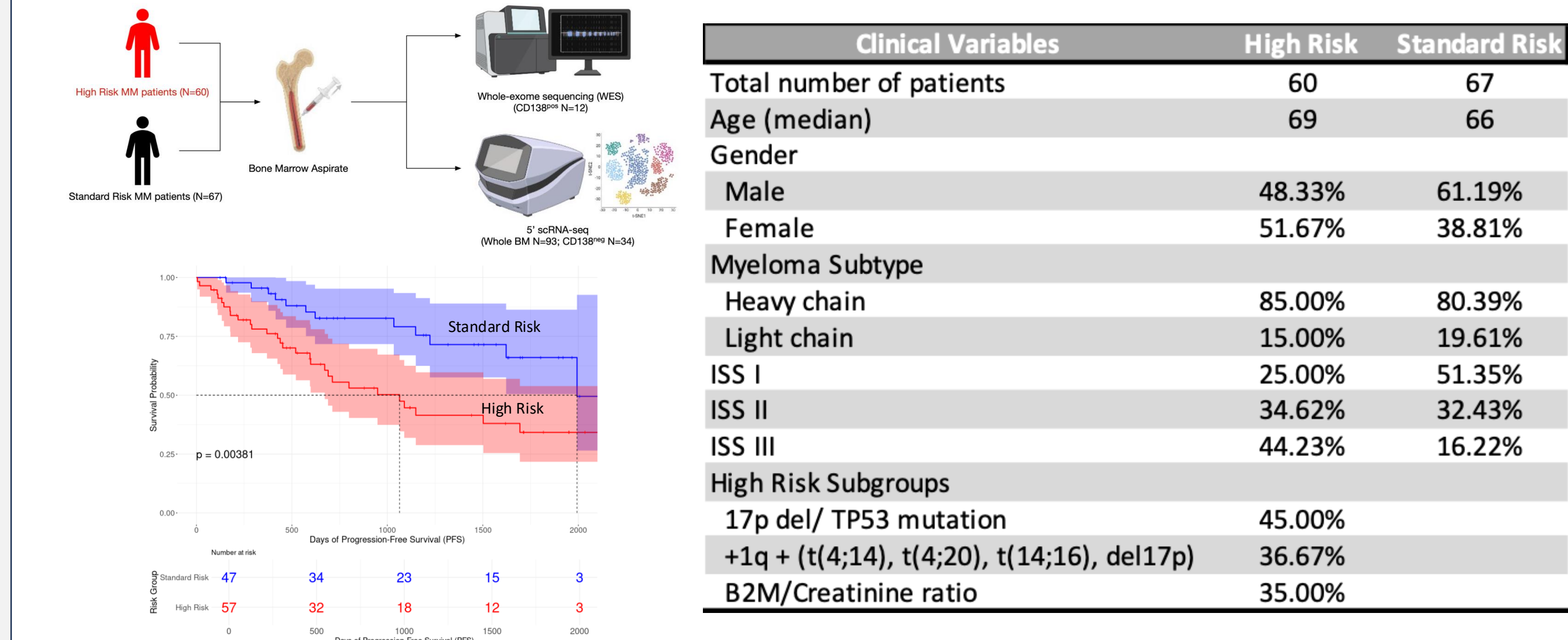
- Bone Marrow samples were collected at diagnosis from 127 NDMM patients, including 60 high-risk (HR) and 67 standard-risk (SR) patients.
- High-risk was defined by the presence of specific alterations:
- HR Group 1:** TP53 deletion (17p13 del) and/or TP53 mutation;
- HR Group 2:**
 - Bi-allelic deletion of 1p (1p32)
 - Gain of 1q (1q21) and deletion of 1p (1p32)
 - Either gain of 1q or deletion of 1p in combination with one of the following translocations: t(4;14) (NSD2), t(14;16) (MAF), or t(14;20) (MAFB)
- HR Group 3:** B2M ≥ 5.5 mg/L with Creatinine < 1.2 mg/dL

Immunophenotyping by 5' single cell RNA-seq

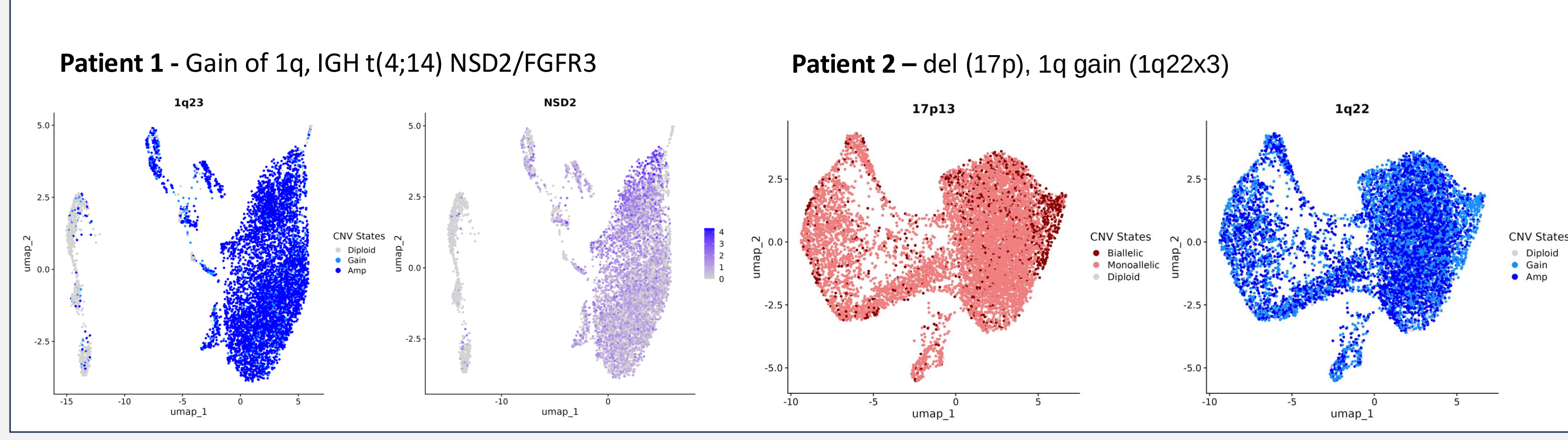
- Single cell RNA sequencing was performed to analyze transcriptional changes in MM cells and in the MM-TME.
- Samples were filtered based on total cell count (> 1000 cells), total unique molecular identifiers (UMIs) count (> 500 UMIs), and mitochondrial gene expression.
- Our dataset was comprised of 879,588 BM cells, categorized into 63,432 myeloid cells, 60,558 NK cells, 409,675 T cells, 47,113 B cells, and 180,292 plasma cells.
- Downstream analysis was performed using R packages Seurat, CellphoneDB, SCENIC, InferCNV and SComatic.

RESULTS

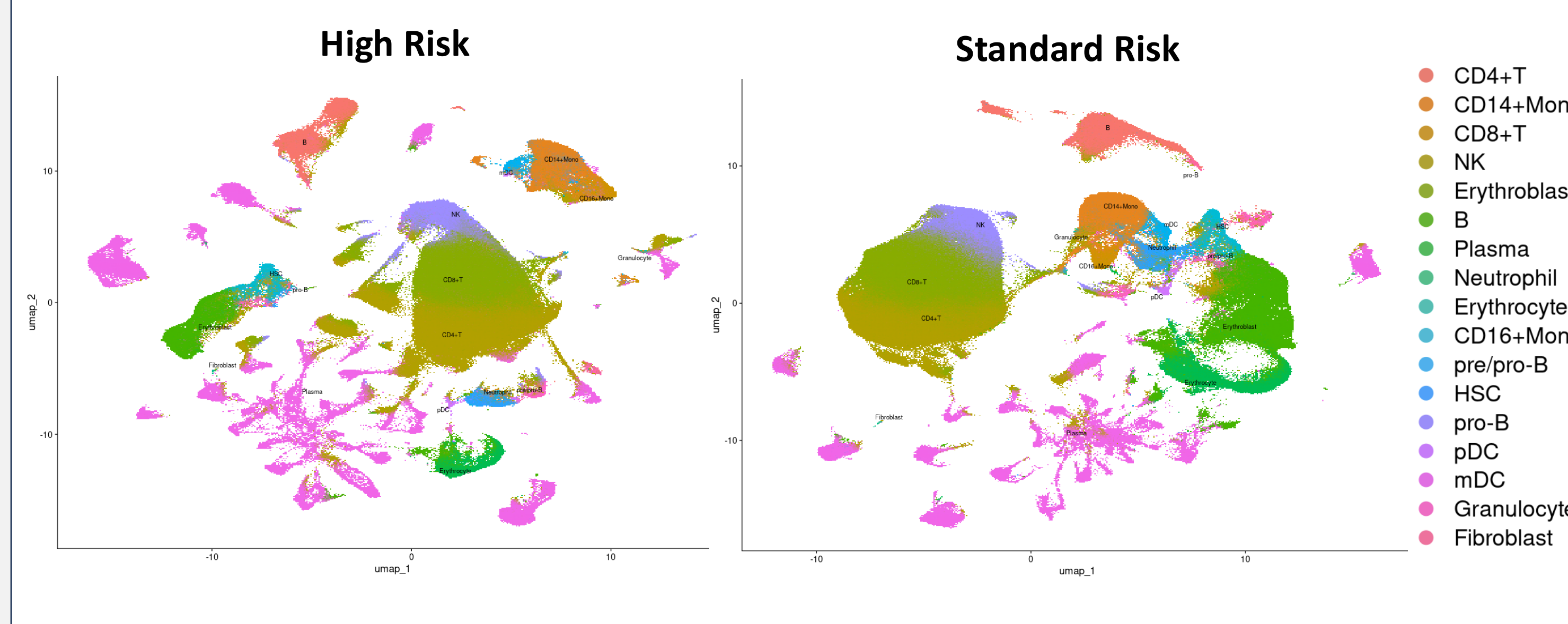
1. Study Design Schematic and Table of Patient Characteristics



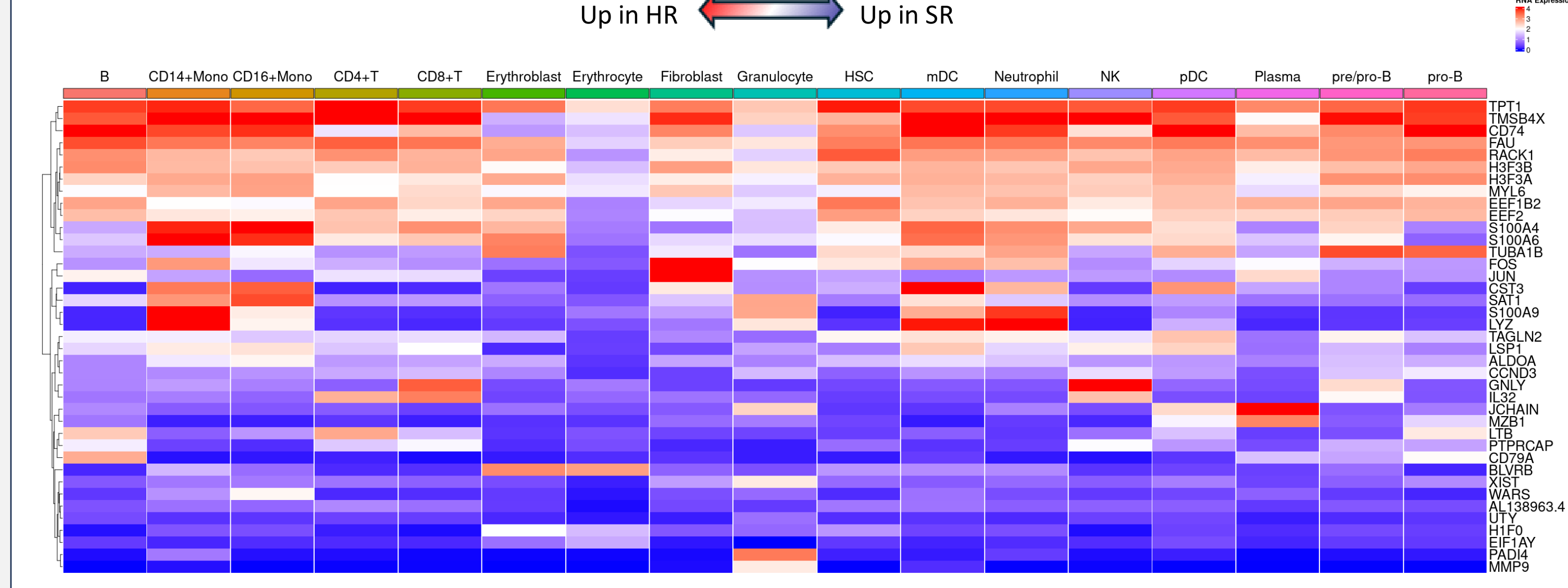
2. High-Risk Lesions are Identified at the Single-Cell Level through Integrated Tumor Analysis



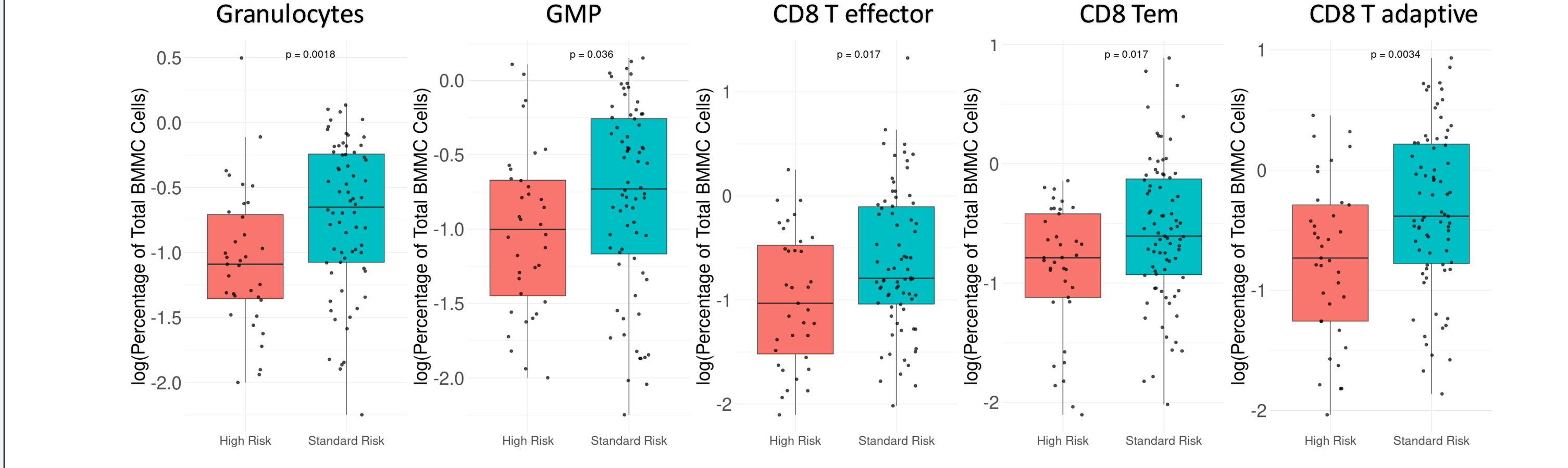
3. Single Cells are Categorized into Distinct Clusters Based on Their Gene Expression Profiles



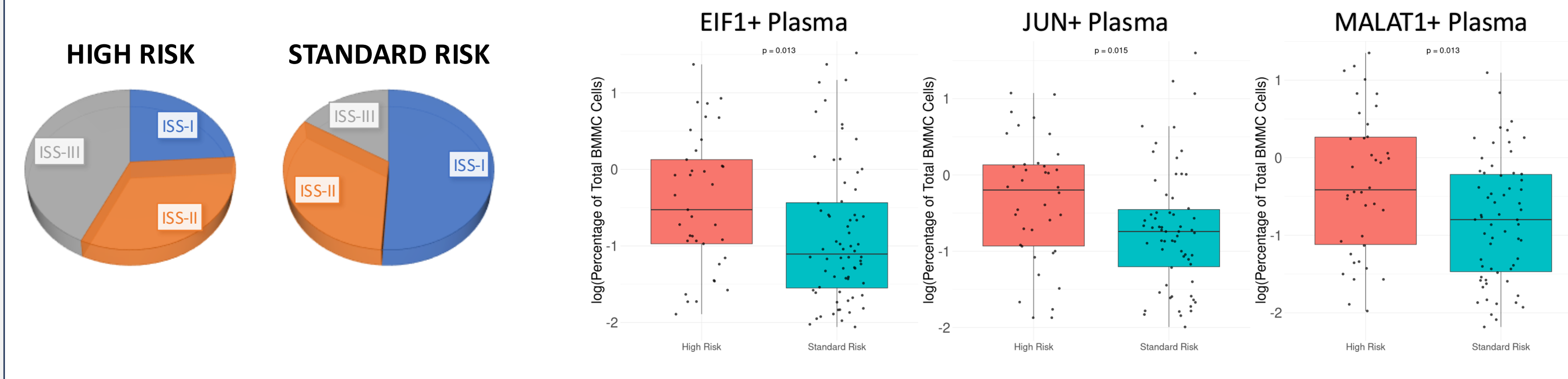
4. Pseudobulk Differential Gene Expression Analysis Reveals Distinct Genes Enriched in High-Risk Versus Standard-Risk Cohorts



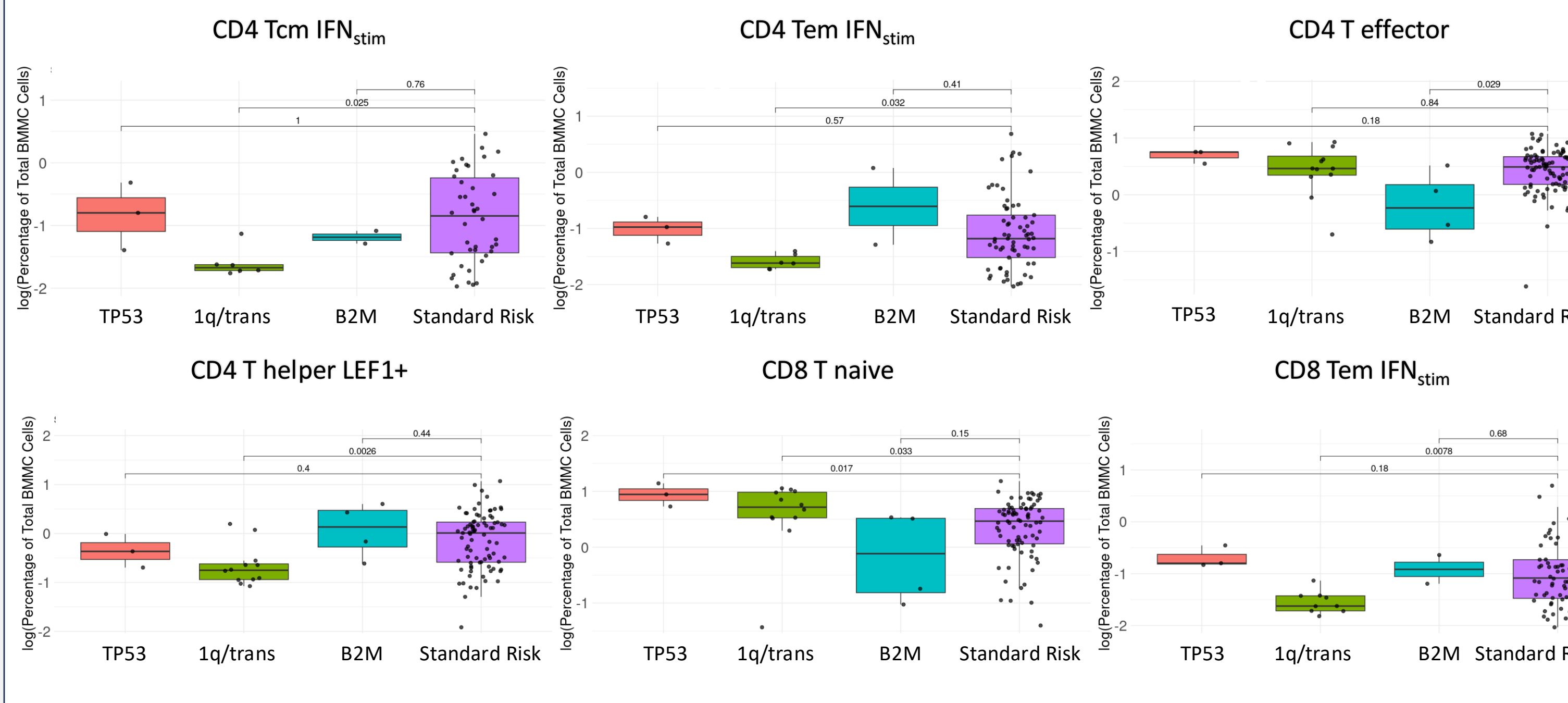
5. Standard Risk Patients Have Significantly Higher Frequencies of Granulocytes, Granulocyte-Monocyte Progenitors (GMP) and CD8 T Cell Subgroups Compared to High Risk



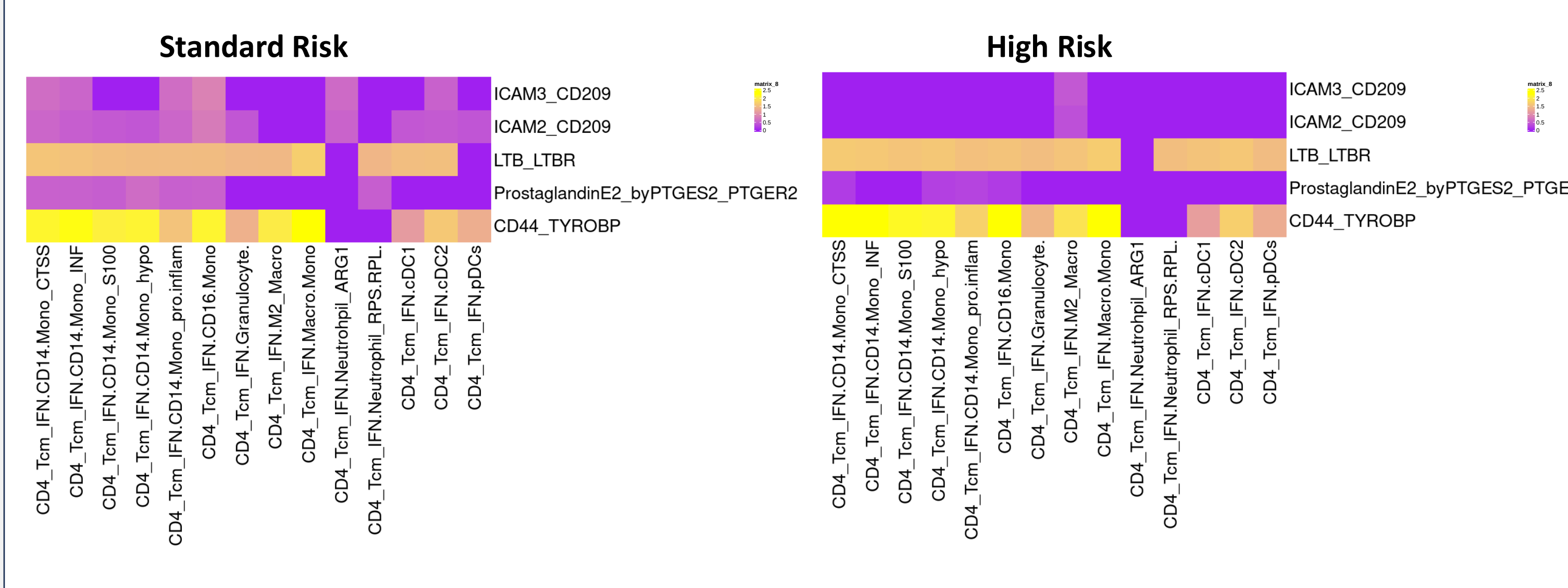
6. High Risk Patients Had Significantly Higher Frequencies of Plasma cell Subsets When Compared to Standard Risk, which Also Correlated with Higher Proportion of Patients with ISS Stage 3



7. High-Risk Groups Exhibit Reduced Frequencies of Key T Cell Subtypes, Indicating Compromised Immune Surveillance and Effector Function



8. Stronger APC – T Cell Interactions Related to Leukocyte Recruitment, T Cell Activation, IL-2 and IFN-γ Production are Observed in Standard Risk Compared to High Risk



CONCLUSIONS

- Our findings indicate that HR patients have a lower frequency of myeloid subpopulations (granulocytes and granulocyte-monocyte progenitors (GMP)) and a higher frequency of plasma cell subsets compared to SR patients (p < 0.05), suggesting a potential depletion of antigen-presenting cells (APCs), possibly due to increased tumor burden.
- HR group 2 had significantly lower frequencies of IFN-stimulated CD4 and CD8 T cell populations compared to SR, suggesting that gain of 1q associated with deletion of 1p or t(4;14) / t(14;16) / t(14;20) may contribute to dysregulation of IFN-related T cell response.

CLINICAL RELEVANCE

- Understanding the interplay between high-risk alterations and the MM-TME can improve patient risk stratification and lead to better treatment outcomes for HR MM.
- Single-cell analysis identified a depletion of Antigen-Presenting Cell (APC) populations and CD8 T cell subsets in the HR Cohort, which may result in a less effective cell-mediated immune response in HR MM.

ACKNOWLEDGEMENTS

Mount Sinai Tisch Cancer Center

MULTIPLE MYELOMA Research Foundation

We would also like to thank all the patients and their caregivers.